Physics 3221 Spring 2026: Midterm 2

Grade:

1. A Strange Force (20pts)

A particle travels between points A and B as shown in Fig. 1. The particle is subject to the following force:

$$\vec{F} = k\,y\hat{x} + l\frac{dy}{dx}\hat{y}\,,$$

where k and l are constants.

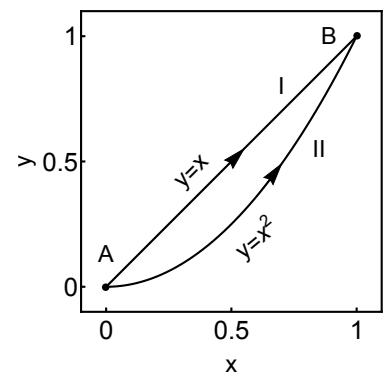

Figure 1: Points A and B connected by various paths

(a) **[5 pts]** When $l = 0$, is this force path-independent? Verify by calculating its curl $\vec{\nabla} \times \vec{F}$.

$(\vec{\nabla}\times\vec{F})_z = 0 - k \neq 0 \quad \Rightarrow$ not path-independent

(b) **[10 pts]** Calculate the work $\vec{F}$ does on the particle if it travels between point A and point B along path I given by $y = x$

$\frac{dy}{dx} = 1, \quad dy = dx$

$$W = \int \vec{F}\cdot d\vec{r} = \int ky\,dx + l\frac{dy}{dx}\,dy$$

$$= \int_0^1 kx\,dx + \int_0^1 l(1)\,dx = \frac{k}{2}x^2\Big|_0^1 + lx\Big|_0^1 = \boxed{\frac{k}{2} + l = W}$$

(c) **[5 pts]** Calculate the work $\vec{F}$ does on the particle if it travels between point A and point B along path II given by $y = x^2$

$\frac{dy}{dx} = 2x \qquad dy = 2x\,dx$

$$W = \int ky\,dx + l(2x)\,dy$$

$$W = \int_0^1 k\,x^2 dx + 2xl\;2x\,dx$$

$$W = \int_0^1 kx^2 dx + \int_0^1 4lx^2 dx$$

$$W = (k+4l)\left(x^3/3\right)_0^1$$

$$\boxed{W = \frac{k+4l}{3}}$$

Grade:

2. The Damped Harmonic Oscillator (20 Points)

A mass attached to a spring as depicted in Fig. 2 experiences only the following forces:

$$\text{Spring force:} \quad \vec{F}_s = -k\vec{x}$$
$$\text{Drag force:} \quad \vec{F}_d = -b\vec{v}$$

where $\vec{x}$ is defined such that at $\vec{x} = 0$, the spring is at its equilibrium position. All motion of the spring occurs along the x-axis.

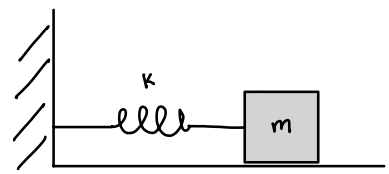

Figure 2: A mass m attached to a spring with spring constant k, moving only in the x-direction

(a) **[7 pts]** Use Newton's 2nd Law to write down a differential equation describing the motion of the spring, and solve it to find the general solution $x(t)$ for the spring. *Note: I have not yet given you initial conditions, so you can leave two undetermined constants in your answer.*

$m\ddot{x} = -kx - b\dot{x}$

$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$,

$\boxed{2\beta \equiv b/m \quad \omega_0^2 \equiv k/m}$

$x = e^{rt} \Rightarrow r^2 e^{rt} + 2\beta r e^{rt} + \omega_0^2 e^{rt} = 0$

$r^2 + 2\beta r + \omega_0^2 = 0$

$\Rightarrow r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$

$\Rightarrow \boxed{x(t) = c_1 e^{(-\beta + \sqrt{\beta^2 - \omega_0^2})t} + c_2 e^{(-\beta - \sqrt{\beta^2 - \omega_0^2})t}}$

(b) **[6 pts]** If $k = b^2/m$, is the system oscillating? Why or why?

The system oscillates when $\beta^2 - \omega_0^2 < 0$ (imaginary # in exponent)

$\Rightarrow \beta^2 < \omega_0^2$

$\Rightarrow \left(\frac{b}{2m}\right)^2 < \frac{k}{m}$

$\frac{b^2}{4m^2} < \frac{k}{m} \Rightarrow \frac{b^2}{4m} < k$

If $k = \frac{b^2}{m}$

then $\frac{b^2}{4m} = \frac{k}{4} < k$

$\boxed{\text{so the system oscillates}}$

THE REST OF PROBLEM 2 IS ON THE NEXT PAGE

(c) **[7 pts]** Assume $k = b^2/m$. At $t = 0$, assume the mass is released from rest at position x_0. At time t_1, the mass is at a local maximum and turning back around towards equilibrium, where t_1 is given by:

$$t_1 = 2\pi/\omega_1\,, \qquad \omega_1 = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}\,.$$

$$\omega_1 = \sqrt{\frac{b^2}{m^2} - \frac{b^2}{4m^2}} = \frac{\sqrt{3}b}{2m}$$

How much work has been done by $\vec{F}_d$ from $t = 0$ to $t = t_1$? You will get full credit for obtaining the correct magnitude of work (regardless of sign).

$$x(t) = c_1 e^{(-\beta + \sqrt{\beta^2 + \omega_0^2})t} + c_2 e^{(-\beta - \sqrt{\beta^2 + \omega_0^2})t} \qquad \omega_1^2 = \omega_0^2 - \beta^2$$

$$= e^{-\beta t}\left(c_1 e^{i\omega_1 t} + c_2 e^{-i\omega_1 t}\right)$$

$$x_0 = c_1 + c_2 \qquad x_1 = e^{-\beta t_1}\left(c_1 e^{2\pi i} + c_2 e^{-2\pi i}\right)$$

$$x_1 = e^{-2\pi\beta/\omega_1}(c_1 + c_2) = x_0 e^{-2\pi\beta/\omega_1}$$

$$\Delta E = \Delta T + \Delta U = W_{drag}$$

$$0 + \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_0^2 = W_{drag}$$

$$\tfrac{1}{2}kx_0^2 e^{-4\pi\beta/\omega_1} - \tfrac{1}{2}kx_0^2 = W_{drag}$$

$$\boxed{\tfrac{1}{2}kx_0^2\left(e^{-\frac{2\pi b}{m\omega_1}} - 1\right) = W_{drag}}$$

$$\boxed{\frac{b^2x_0^2}{2m}\left(e^{-\frac{4\pi}{\sqrt{3}}} - 1\right) = W_{drag}}$$

Grade:

3. Driven Harmonic Oscillator (20 Points)

Consider now the mass attached to a spring as shown in Fig. 2 being acted on by a periodic driving force $F(t)$. A plot for $f(t)$ is given in Fig. 3, where $f(t) \equiv F(t)/m$. Below, fill in the blanks for the following derivation for the solution $x(t)$ that gives the resulting motion of the mass.

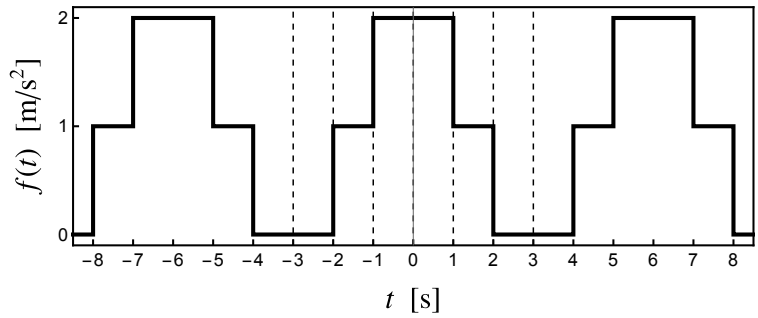

Figure 3: The function $f(t)$ related to the driving force $F(t)$. As shown in the plot, $f(t)$ has a period of 6 s.

The equation of motion for this mass on a spring is:

$$\ddot{x}(t) + 2\beta\dot{x}(t) + \omega_0^2 x(t) = f(t)$$

(a) **[3 pts]** Find β and ω_0.

$2\beta = b/m$ $\qquad$ $\omega_0^2 = k/m$

$\Rightarrow \boxed{\beta = \frac{b}{2m} \qquad \omega_0 = \sqrt{k/m}}$

Next, use the Fourier expansion for $f(t)$:

$$f(t) = \sum_n a_n \cos(n\omega t) + b_n \sin(n\omega t)$$

(b) **[4 pts]** Find a_0

$$a_0 = \frac{2}{\tau}\int_0^{(1s)} 2\,(m/s^2)\,dt + \frac{2}{\tau}\int_{(1s)}^{(2s)} 1\,(m/s^2)\,dt = \frac{4}{\tau}(m/s^2)\,t\Big|_0^{1s} + \frac{2}{\tau}(m/s^2)\,t\Big|_{1s}^{2s}$$

$$= \frac{4}{\tau}\,m/s^2\,(1s) + \frac{2}{\tau}\,m/s^2\,(1s) = \frac{6}{\tau}\,m/s = \frac{6}{6s}\,m/s = \boxed{1\,m/s^2}$$

THE REST OF PROBLEM 3 IS ON THE NEXT PAGE

(c) **[4 pts]** Find a_n for $n > 0$.

See next page

(d) **[4 pts]** Find b_n for general n. $b_n = 0$

(e) **[1 pt]** Given a_0, a_n $(n > 0)$, and b_n, write $f(t)$ as a Fourier expansion:

$$f(t) = \underline{\frac{6\,\text{m/s}}{\tau}} + \sum_{n=1}^{\infty} \underline{\frac{2}{n\pi}\left[\sin n\omega + \sin 2n\omega\right] \cos n\omega t \quad [\text{m/s}^2]}$$

Now we have $f(t)$ as a sum of $f_n(t)$ functions, so we can write the general solution to the equation of motion $x(t)$ as:

$$f(t) = \sum_{n=0}^{\infty} f_n(t) \qquad \Rightarrow \qquad x(t) = \sum_{n=0}^{\infty} x_n(t)\,,$$

where each $x_n(t)$ solves the following differential equation:

$$\ddot{x}_n(t) + 2\beta\dot{x}_n(t) + \omega_0^2 x_n(t) = f_n(t)$$

We also know the following differential equations have the following solutions:

$$\ddot{y}_1(t) + 2\beta\dot{y}_1(t) + \omega_0^2 y_2(t) = f_0 \cos\omega t \qquad \Rightarrow \qquad y_1(t) = A\cos(\omega t - \delta)$$
$$\ddot{y}_2(t) + 2\beta\dot{y}_2(t) + \omega_0^2 y_2(t) = f_0 \sin\omega t \qquad \Rightarrow \qquad y_2(t) = A\sin(\omega t - \delta)$$

where A and δ are on the equation sheet.

(f) **[3 pts]** Using the information above, find $x_n(t)$

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3c

$$a_n = \frac{4}{\tau}\int_0^{(1s)} 2\left(m/s^2\right)\cos n\omega t\,dt + \frac{4}{\tau}\int_{(1s)}^{(2s)}\left(1\,m/s^2\right)\cos n\omega t\,dt = \frac{4}{\tau}\left(2\,m/s^2\right)\frac{1}{n\omega}\sin n\omega t\Big|_0^{1s} + \frac{4}{\tau}\left(1\,m/s^2\right)\frac{1}{n\omega}\sin n\omega t\Big|_{1s}^{2s}$$

$$= \frac{8}{n\omega\tau}\left(m/s^2\right)\sin\left[n\omega(1s)\right] - 0 + \frac{4}{n\omega\tau}\left(m/s^2\right)\sin\left[n\omega(2s)\right] - \frac{4}{n\omega\tau}\left(m/s^2\right)\sin\left[n\omega(1s)\right]$$

Note: $\omega\tau = 2\pi$ & $\omega = \frac{2\pi}{\tau} = \frac{2\pi}{6s} = \frac{\pi}{3}\,s^{-1}$

$$a_n = \frac{8}{2\pi n}\left(m/s^2\right)\sin\left(n\pi/3\right) + \frac{4}{2\pi n}\left(m/s^2\right)\sin\left(\frac{2n\pi}{3}\right) - \frac{4}{2\pi n}\left(m/s^2\right)\sin\left(n\pi/3\right)$$

$$\boxed{a_n = \frac{2}{\pi n}\sin\left(n\pi/3\right) + \frac{2}{\pi n}\sin\left(2n\pi/3\right) \quad m/s^2}$$

$$a_n = \frac{2}{n\pi}\sin\left(2n\omega\right) + \frac{2}{n\pi}\sin\left(n\omega\right) \quad m/s^2$$

3f

$$\ddot{x}_n + 2\beta\dot{x}_n + \omega_0^2 x_n = a_n\cos n\omega t$$

$$\Rightarrow \boxed{x_n = A_n\cos\left(n\omega t - \delta_n\right)} \quad n>0$$

$$A_n^2 = \frac{a_n^2}{\left(\omega_0^2 - n^2\omega^2\right)^2 + 4\beta^2 n^2\omega^2} \Rightarrow \boxed{A_n = \frac{\frac{2}{n\pi}\left(\sin 2n\omega + \sin n\omega\right)}{\sqrt{\left(\omega_0^2 - n^2\omega^2\right)^2 + 4\beta^2 n^2\omega^2}}}$$

$$\omega = \frac{2\pi}{\tau} = \pi/3\ s^{-1}$$

$$\boxed{\delta_n = \tan^{-1}\left(\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)}$$

$n>0$

For $n=0$:

$$\ddot{x}_0 + 2\beta\dot{x}_0 + \omega_0 x_0 = a_0$$

$$\left.\begin{aligned} x_0(t) &= a_0/\omega_0 \\ \dot{x}_0(t) &= 0 \\ \ddot{x}_0(t) &= 0 \end{aligned}\right\} \Rightarrow x_0 = \text{solution}$$

$$\boxed{x_0 = a_0/\omega_0^2} = \frac{6\ m/s}{\tau\,\omega_0^2}$$

$n=0$

(g) **[1 pt]** Add the x_n solutions together to get the full solution from the motion:

$$x(t) = \sum_{n=0}^{\infty} x_n(t) = \ ?$$

$$x(t) = \frac{6\,m/s}{\tau\,\omega_0^2} + \sum_{n=1}^{\infty} A_n \cos(n\omega t - \delta_n) + \underbrace{C_1 e^{(-\beta + \sqrt{\beta^2 - \omega_0^2})t} + C_2 e^{(-\beta - \sqrt{\beta^2 - \omega_0^2})t}}_{\text{homogeneous solution}}$$

Where A_n, δ_n are: from previous part

$$\boxed{A_n = \frac{\frac{2}{n\pi}(\sin 2n\omega + \sin n\omega)}{\sqrt{(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2\omega^2}}} \qquad \boxed{\delta_n = \tan^{-1}\left(\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)}$$

Formulas (just for the Midterm)

Newton's 2nd Law

$$\vec{F} = m\ddot{\vec{r}}$$

Cartesian

$$F_x = m\ddot{x} \qquad F_y = m\ddot{y} \qquad F_z = m\ddot{z}$$

Polar

$$F_r = m(\ddot{r} - r\dot{\phi}^2) \qquad F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

Cylindrical

$$F_\rho = m(\ddot{\rho} - \rho\dot{\phi}^2) \quad F_\phi = m(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) \quad F_z = m\ddot{z}$$

Coordinate Systems

Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Cylindrical : $\vec{r} = \rho\hat{\rho} + z\hat{z}$

$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$

Spherical: $\vec{r} = r\hat{r}$

$d\vec{r} = dr\hat{r} + rd\theta\,\hat{\theta} + r^2\sin\theta\, d\phi\,\hat{\phi}$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Fourier Series

$$f(t) = \sum_{n=0}^{\infty}\left[a_n\cos(n\omega t) + b_n\sin(n\omega t)\right]\,,$$

$$\omega = 2\pi/\tau.$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,,$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n > 0$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,,$$

Hook's Law & Simple Harmonic Motion

$$F = -kx \Leftrightarrow U = \frac{1}{2}kx^2$$

$$x(t) = A\cos(\omega t - \delta)$$

Driven Oscillator

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2)}$$

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Work

$$W = \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,') \cdot d\vec{r}\,'$$

Conservative forces

$$U(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,') \cdot d\vec{r}\,'$$

$$F = -\vec{\nabla}U$$

$$\vec{\nabla} \times \vec{F} = 0$$

Useful Integrals

$$\int x\sin(ax)dx = -\frac{x}{a}\cos(ax) + \frac{1}{a^2}\sin(ax)$$

$$\int x\cos(ax)dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

Gradient and Curl

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\vec{\nabla}f = \hat{\rho}\frac{\partial f}{\partial r\rho} + \hat{\theta}\frac{1}{\rho}\frac{\partial f}{\partial \theta} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla} \times \vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$